



Conventional ML Methods

Applied Machine Learning

Explainable AI

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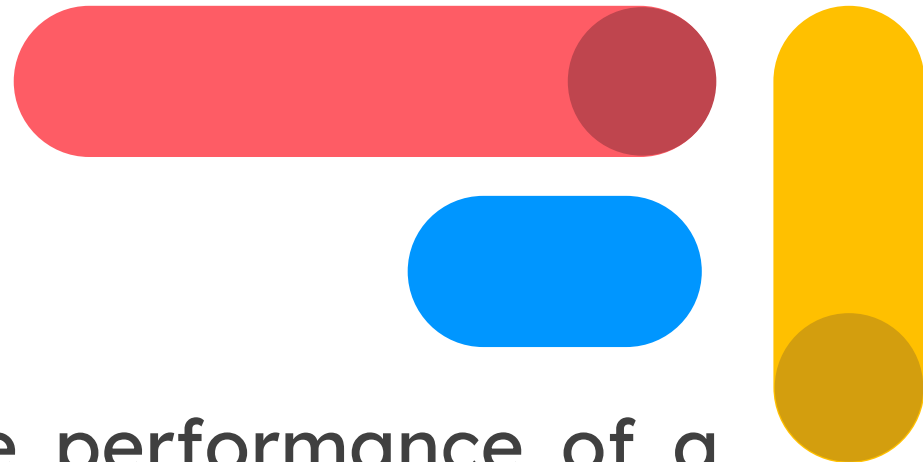


Outline

#GeoAI #Mobility #Geography #STDS

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- 04 Partial Dependence & Individual Conditional Expectation
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Model Explanation



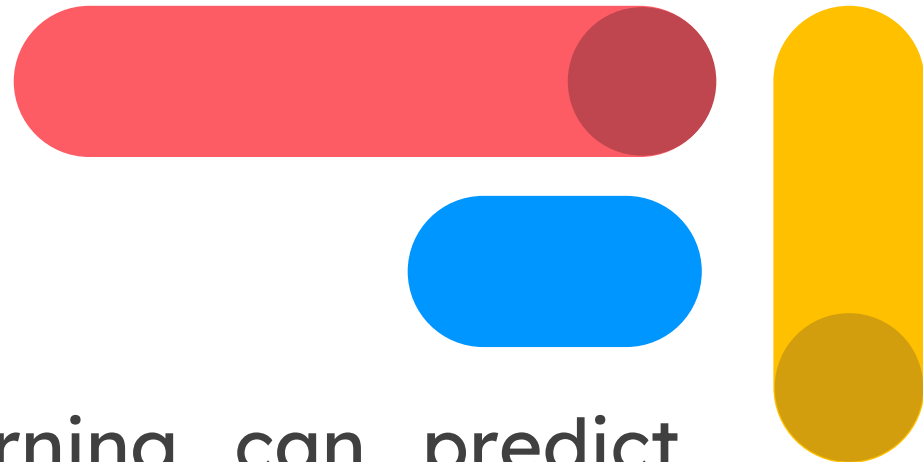
When you trained and evaluate the performance of a model, you may ask what is the next?

The answer is you may try to explain what is going on in this black box (model itself)?

We will guide you to learn two general solutions for abovementioned issues:

- a) feature importance
- b) explainable AI

Model Explanation



From the beginning, machine learning can predict accurate results, but what we have is ...

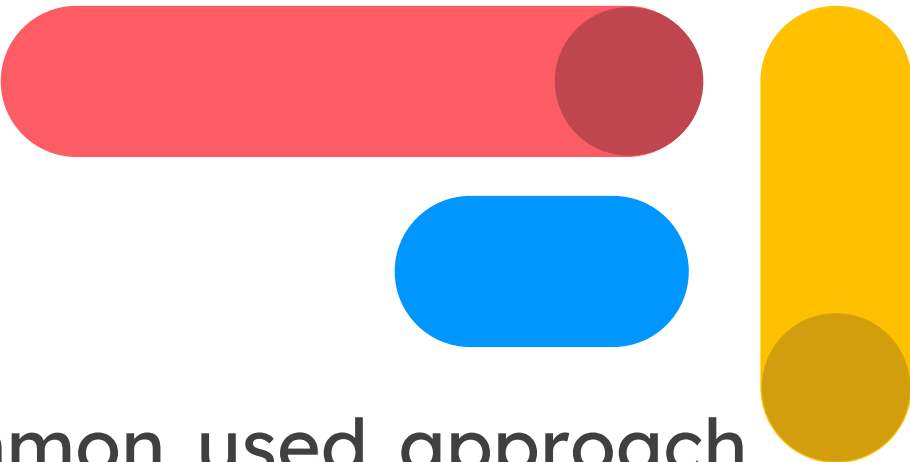
$$\hat{y} = f(X)$$

What we really care about ...

- 1) Which features drive the model?
- 2) How does each feature affect the prediction?
- 3) Why does the model make a particular prediction?
- 4) Can we trust the model's decision?

Core issue :: What has the model learn?

Feature Importance



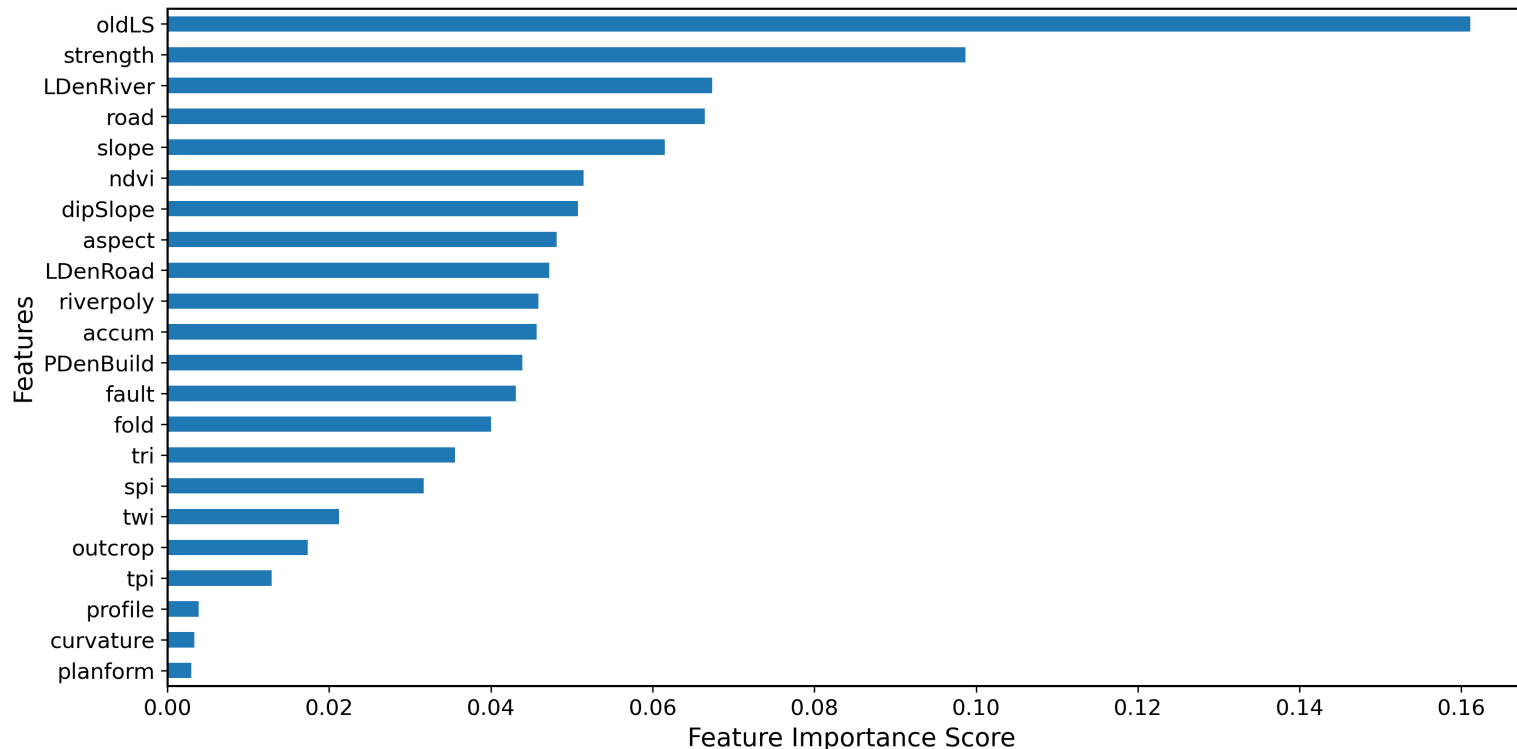
Feature importance is the most common used approach to know your model structure and directly computed by scikit learn function.

Q :: Which features contributes most to the model?

Feature importance quantifies how strongly a trained model relies on different features when making predictions?

Feature Importance

For example, we have a feature importance as follows.

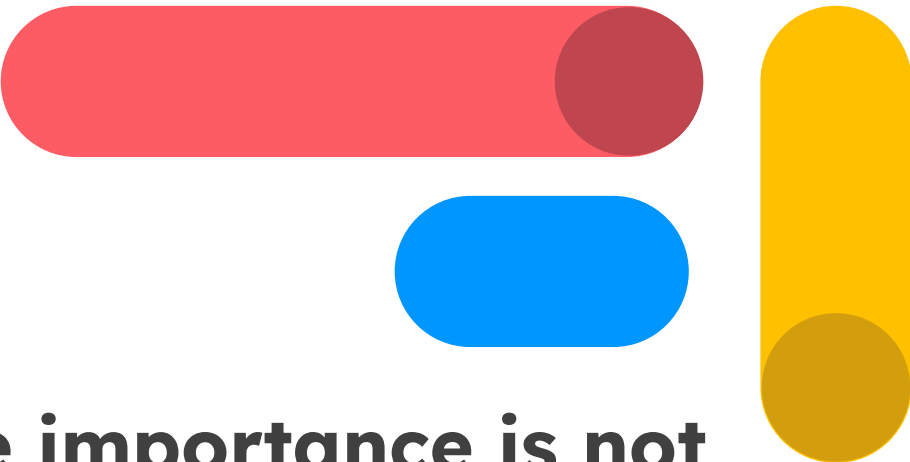


Chan et al. (2025) Develop an AI Enhanced Data for Landslide Early Warning Model (1/2) (NCDR-S-114042). National Science and Technology Center for Disaster Reduction, New Taipei City, Taiwan.

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Chun-Hsiang Chan (2026)

Feature Importance



First and the most important, **feature importance is not equal to Causal effect**. It just indicates that higher importance means that the model relies more heavily on that feature.

So, how can we know the computation process of feature importance from the previous example?

In the following slides, we demonstrate the calculation of feature importance from scikit learn packages.

Feature Importance

Feature importance function in sklearn will be provided for some of ML models.

Tree-based Models

DecisionTreeClassifier
RandomForestClassifier
ExtraTreesClassifier
AdaBoostClassifier
GradientBoostingClassifier

Gini or Entropy

Linear-based Models

LogisticRegression
LinearSVC
SVC(kernel='linear')

Coefficient (after standardization)

Other Models

KNeighborsClassifier
Naïve Bayes
SVC(kernel='poly')
SVC(kernel='rbf')

Permutation Importance

Feature Importance

For XGBoost, LightGBM, and CatBoost, they have their own feature importance calculation method.

Model	Default	Options	Description
XGBoost	Gain	Weight, Cover	The total objective gain from split/ usage times/ number of covered samples divided by Hessian
LightGBM	Split times	Gain	Split times/ aggregated gain
CatBoost	Prediction Values Changes	Loss Function Change	The average changes in predictions when feature changes

Feature Importance

Feature Importance – Tree-based Models

The feature importance is calculated by impurity reduction.

1) Impurity Reduction at Each Split

$$\Delta I_t = I(t) - \frac{N_L}{N_t} I(t_L) - \frac{N_R}{N_t} I(t_R)$$

$I(t)$: parent impurity; $I(t_L)$: left-child impurity; $I(t_R)$: right-child impurity; N_t : parent samples; N_L, N_R : child samples.

Impurity could be used by Gini index or entropy.

$$Gini = 1 - \sum_k p_k^2$$

Feature Importance

Feature Importance – Tree-based Models

2) Weight by the Number of Samples

We need to further consider the number of training samples of this node to determine the contributions.

$$\text{Contribution}(t) = \frac{N_t}{N} \Delta I_t$$

The nearer split to the root has more impact on model outcome because it covers more observations.

Feature Importance

Feature Importance – Tree-based Models

3) Sum All Splits Using Feature j

If the sample feature has been used for split five times, then we need to aggregate all contributions.

$$FI_j^{raw} = \sum_{t:v(t)=j} \frac{N_t}{N} \Delta I_t$$

Feature Importance

Feature Importance – Tree-based Models

4) Normalization

To facilitate the comparison among all features, we have to do normalization.

$$\therefore FI_j = \frac{FI_j^{raw}}{\sum_k FI_k^{raw}}, \therefore \sum_j FI_j = 1$$

Feature Importance

Permutation Feature Importance

If we use F1-score as primary indicator for model performance, then ...

$$PI_j = \frac{1}{R} \sum_{r=1}^R [Score_{original} - Score_{permuted(j,r)}]$$

R represents the number of shuffle times

Feature Importance

Permutation Feature Importance

If we use F1-score as primary indicator for model performance, then ... F1-score = 0.90

We shuffle one feature order, e.g., Age

Original Age = [25,40,65,12,45]

Shuffled Age = [40,25,12,65,45]

$$PI_{Age} = 0.90 - 0.72 = 0.18$$

$$PI_{SBP} = 0.90 - 0.86 = 0.04$$

$$Age > SBP$$

Partial Dependence & Individual Conditional Expectation

Partial dependence plots (PDP) and individual conditional expectation (ICE) plots can be used to visualize and analyze interaction between the target response and a set of input features of interest.

Partial dependence plots (PDP) show the dependence between the target response and a set of input features of interest, marginalizing over the values of all other input features. Intuitively, we can interpret the partial dependence as the expected target response as a function of the input features of interest.

Partial Dependence & Individual Conditional Expectation

Partial Dependence

Average prediction changes when feature X_j changes

If our model looks like ...

$$\hat{y} = f(X_1, X_2, \dots, X_p)$$

If we want to know the impact of X_j on prediction outcome, ...

$$PD_j(x) = E_{X_{-j}}[f(x, X_{-j})]$$

Therefore, the estimation of real cases,

$$\widehat{PD}_j(x) = \frac{1}{n} \sum_{i=1}^n f(x, x_{i,-j})$$

x : given value to X_j ; $x_{i,-j}$: all features except for X_j in observation i ; n : sample size

Source: https://scikit-learn.org/stable/modules/partial_dependence.html

Partial Dependence & Individual Conditional Expectation

Partial Dependence

If we have a model to predict room rate, we want to know the impact of *Size* on room rate *price*.

$$\begin{aligned} \text{price} &= f(\text{Size}, \text{Age}) \\ \text{given a Size} &= 60 \end{aligned}$$

House	Size	Age
A	50	5
B	70	10
C	100	20

House	Size	Age	Price
A	60	5	520
B	60	10	480
C	60	20	400

$$\begin{aligned} PDP(60) &= \frac{520 + 480 + 400}{3} \\ &= 466.7 \end{aligned}$$

House	Size	Age	Price
A	80	5	680
B	80	10	640
C	80	20	560

$$\begin{aligned} PDP(80) &= \frac{680 + 640 + 560}{3} \\ &= 626.7 \end{aligned}$$

Size	PDP
50	390
60	467
70	547
80	627
90	703
100	780

Size $\uparrow \Rightarrow$ Predicted Price $\uparrow$

Partial Dependence & Individual Conditional Expectation

Individual Conditional Expectation

Similar to a PDP, an individual conditional expectation (ICE) plot shows the dependence between the target function and an input feature of interest. However, unlike a PDP, which shows the average effect of the input feature, an ICE plot visualizes the dependence of the prediction on a feature for each sample separately with one line per sample. Due to the limits of human perception, only one input feature of interest is supported for ICE plots.

Partial Dependence & Individual Conditional Expectation

Individual Conditional Expectation

For observation i ,

$$ICE_i(x) = f(x, x_{i,-j})$$

Given fixed feature of one observation, we observe how the prediction changes when X_j only changes.

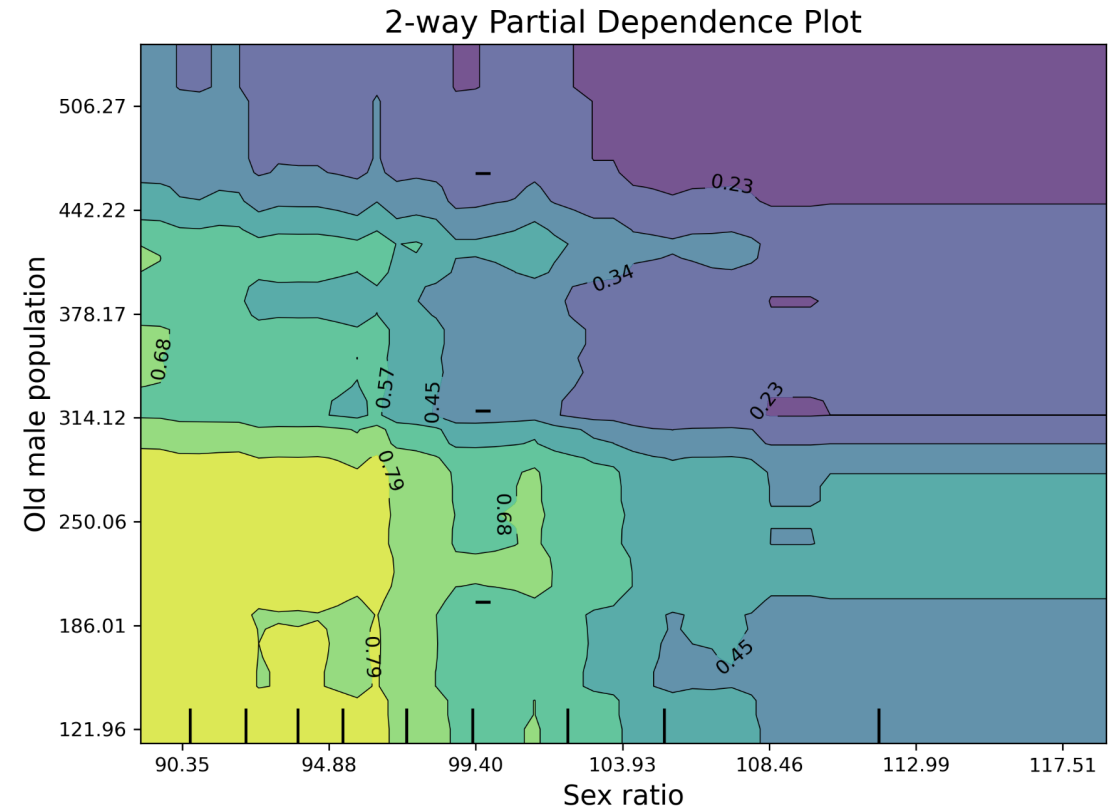
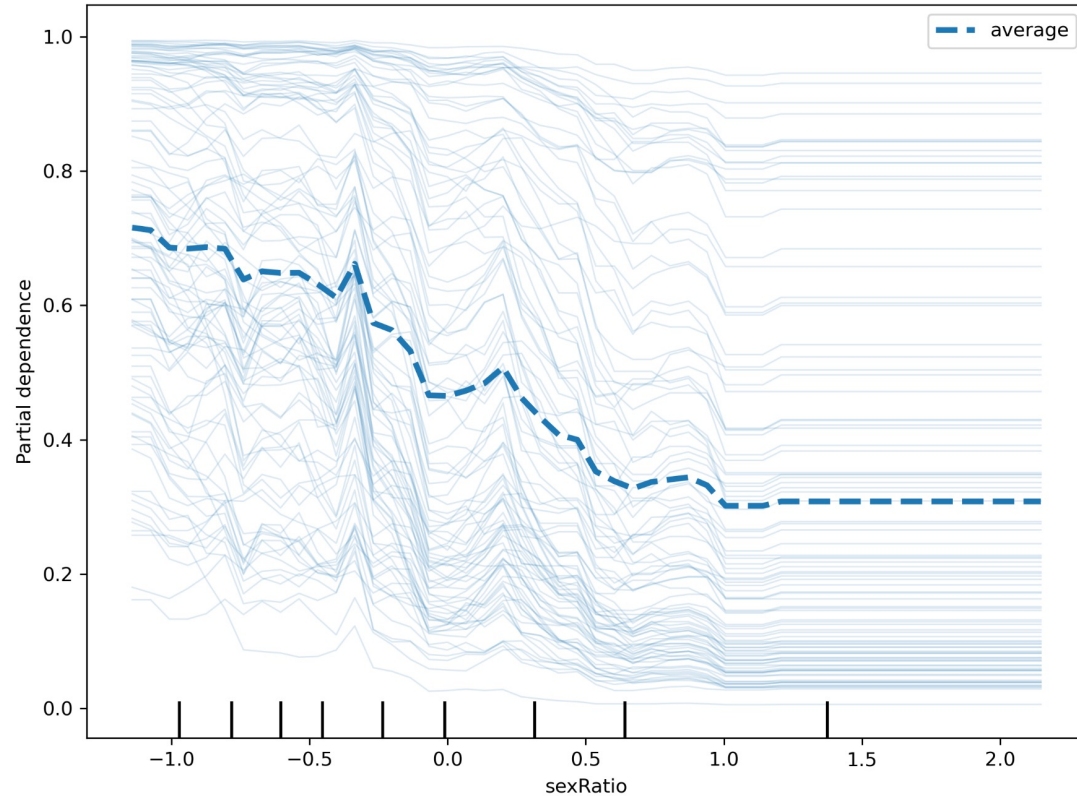
Size	House A	House B	House C
50	440	400	330
60	520	480	400
70	600	560	480
80	680	640	560
90	760	720	640

$$PDP_j = \frac{ICE_A + ICE_B + ICE_C}{3}$$

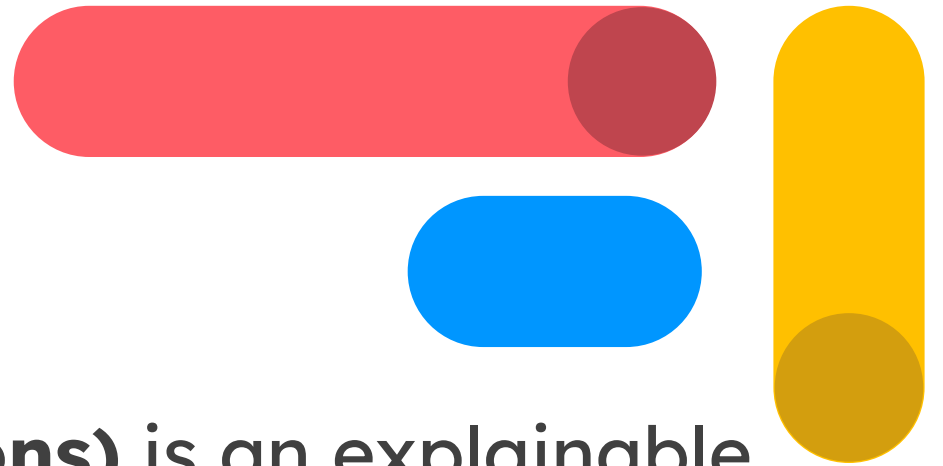
PDP is the average of all ICEs.

Partial Dependence & Individual Conditional Expectation

Individual Conditional Expectation



SHapley Additive exPlanations



SHAP (SHapley Additive exPlanations) is an explainable AI method based on Shapley values from cooperative game theory. It explains a prediction by assigning each feature a contribution value that represents how much that feature pushes the model output away from a baseline prediction.

SHAP explains how much each feature contributes to a specific model prediction.

It is mainly a local explanation method, but SHAP values can also be aggregated across observations to provide global feature importance and overall model interpretation.

SHapley Additive exPlanations

What does SHAP want to do?

If we want to predict a person whether have one disease, then here ...

Feature	Value
Age	65
BMI	30
Exercise	1 hr/week
Smoking	Yes

$$P(Y = 1) = 0.80$$

SHAP divides into several parts for showing the contribution of each feature.

Prediction = Baseline + Feature contributions

$$0.80 = \underbrace{0.40}_{\text{baseline}} + \underbrace{0.18}_{\text{Age}} + \underbrace{0.10}_{\text{BMI}} + \underbrace{(-0.03)}_{\text{Exercise}} + \underbrace{0.15}_{\text{Smoking}}$$

SHapley Additive exPlanations

SHAP utilizes an additive explanation model.

$$f(x) = E[f(X)] + \sum_{j=1}^p \phi_j$$

$E[f(X)]$ represents baseline/ expected value; ϕ_j is the SHAP value of the j -th feature.

So, if $\phi_j > 0$ represents positive impact on prediction from this feature j ; $\phi_j < 0$ is negative impact on prediction.

$$\text{expected value} + \sum_j SHAP_j = f(x)$$

SHapley Additive exPlanations

How to compute Shapley value?

$$\phi_j = \sum_{S \subseteq \{x_1, \dots, x_p\} \setminus \{x_j\}} \underbrace{\frac{|S|! (p - |S| - 1)!}{p!}}_{\text{The probability of this combination}} \underbrace{[f(S \cup \{x_j\}) - f(S)]}_{\text{Contribution of } x_j}$$

$$f_x(S) = \int \hat{f}(x_1, \dots, x_p) d\mathbb{P}_{x \notin S} - E_x \left(\hat{f}(X) \right)$$

$x_1, \dots, x_p$: all features except for feature x_j

p : number of features

S : the subset of all features except for feature x_j

$f(S)$: the prediction value of subset S substrate with expected value (average prediction)

SHapley Additive exPlanations

How to compute Shapley value?

$$\phi_j = \sum_{S \subseteq \{x_1, \dots, x_p\} \setminus \{x_j\}} \frac{|S|! (p - |S| - 1)!}{p!} [f(S \cup \{x_j\}) - f(S)]$$

Given a set of features $\{A, B, C, D\}$, we have to compute the Shapley value of D .

Number of features: $p = 4$

All features except for D is $\{A, B, C\}$

All possibility of coalition S is $2^3 = 8$, i.e.,

$\emptyset, \{A\}, \{B\}, \{C\}, \{A, B\}, \{B, C\}, \{A, C\}, \{A, B, C\}$

SHapley Additive exPlanations

How to compute Shapley value?

$$\phi_j = \sum_{S \subseteq \{x_1, \dots, x_p\} \setminus \{x_j\}} \frac{|S|! (p - |S| - 1)!}{p!} [f(S \cup \{x_j\}) - f(S)] \quad \{\text{D, A, B, C}\}$$

1) D is the 1st important, then S is $\emptyset$ and contribution will be $f(D) - f(\emptyset)$

$$\text{Probability} = \frac{0! (4 - 0 - 1)!}{4!} = \frac{0! 3!}{4!} = \frac{6}{24} = \frac{1}{4}$$

All possibilities of D in the 1st place are $C_0^3 = 1$ combination.

SHapley Additive exPlanations

How to compute Shapley value?

$$\phi_j = \sum_{S \subseteq \{x_1, \dots, x_p\} \setminus \{x_j\}} \frac{|S|! (p - |S| - 1)!}{p!} [f(S \cup \{x_j\}) - f(S)] \quad \{\text{A, D, B, C}\}$$

2) D is the 2nd important, then S is 1 and contribution will be $f(A, D) - f(A)$

$$\text{Probability} = \frac{1! (4 - 1 - 1)!}{4!} = \frac{1! 2!}{4!} = \frac{2}{24} = \frac{1}{12}$$

All possibilities of D in the 2nd place are $C_1^3 = 3$ combinations.

SHapley Additive exPlanations

How to compute Shapley value?

$$\phi_j = \sum_{S \subseteq \{x_1, \dots, x_p\} \setminus \{x_j\}} \frac{|S|! (p - |S| - 1)!}{p!} [f(S \cup \{x_j\}) - f(S)] \quad \{\mathbf{A}, \mathbf{B}, \mathbf{D}, \mathbf{C}\}$$

3) D is the 3rd important, then S is 2 and contribution will be $f(A, B, D) - f(A, B)$

$$\text{Probability} = \frac{2! (4 - 2 - 1)!}{4!} = \frac{2! 1!}{4!} = \frac{2}{24} = \frac{1}{12}$$

All possibilities of D in the 3rd place are $C_2^3 = 3$ combinations.

SHapley Additive exPlanations

How to compute Shapley value?

$$\phi_j = \sum_{S \subseteq \{x_1, \dots, x_p\} \setminus \{x_j\}} \frac{|S|! (p - |S| - 1)!}{p!} [f(S \cup \{x_j\}) - f(S)] \quad \{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}\}$$

4) D is the 4th important, then S is 3 and contribution will be $f(A, B, C, D) - f(A, B, C)$

$$\text{Probability} = \frac{3! (4 - 3 - 1)!}{4!} = \frac{3! 0!}{4!} = \frac{6}{24} = \frac{1}{4}$$

All possibilities of D in the 4th place are $C_3^3 = 1$ combinations.

SHapley Additive exPlanations

How to compute Shapley value?

$$\phi_j = \sum_{S \subseteq \{x_1, \dots, x_p\} \setminus \{x_j\}} \frac{|S|! (p - |S| - 1)!}{p!} [f(S \cup \{x_j\}) - f(S)]$$

$$\begin{aligned} \phi_D = & \frac{1}{4} [f(D) - f(\emptyset)] + \\ & \frac{1}{12} [f(A, D) - f(A)] + \frac{1}{12} [f(B, D) - f(B)] + \frac{1}{12} [f(C, D) - f(C)] + \\ & \frac{1}{12} [f(A, B, D) - f(A, B)] + \frac{1}{12} [f(B, C, D) - f(B, C)] + \frac{1}{12} [f(A, C, D) - f(A, C)] + \\ & \frac{1}{4} [f(A, B, C, D) - f(A, B, C)] \end{aligned}$$

SHapley Additive exPlanations

How to compute Shapley value?

$$f_x(S) = \int \hat{f}(x_1, \dots, x_p) d\mathbb{P}_{x \notin S} - E_x(\hat{f}(X))$$

$$x = (A = 10, B = 20, C = 30, D = 40); S = \{A, B\} \Rightarrow f_x(\{A, B\})$$

Given A and B as 10 and 20; C and D are unknown.

$$E_{C,D}[\hat{f}(A = 10, B = 20, C, D)]$$

C	D	$\hat{f}(A = 10, B = 20, C, D)$
10	10	0.30
10	20	0.40
20	10	0.50
20	20	0.60

$$\begin{aligned} E_{C,D}[\hat{f}(A = 10, B = 20, C, D)] \\ = \frac{0.3 + 0.4 + 0.5 + 0.6}{4} = 0.45 \end{aligned}$$

SHapley Additive exPlanations

How to compute Shapley value?

$$f_x(S) = \int \hat{f}(x_1, \dots, x_p) d\mathbb{P}_{x \notin S} - E_x \left(\hat{f}(X) \right)$$

$E_x \left(\hat{f}(X) \right)$ is the baseline prediction

$$E_x \left(\hat{f}(X) \right) = 0.30$$

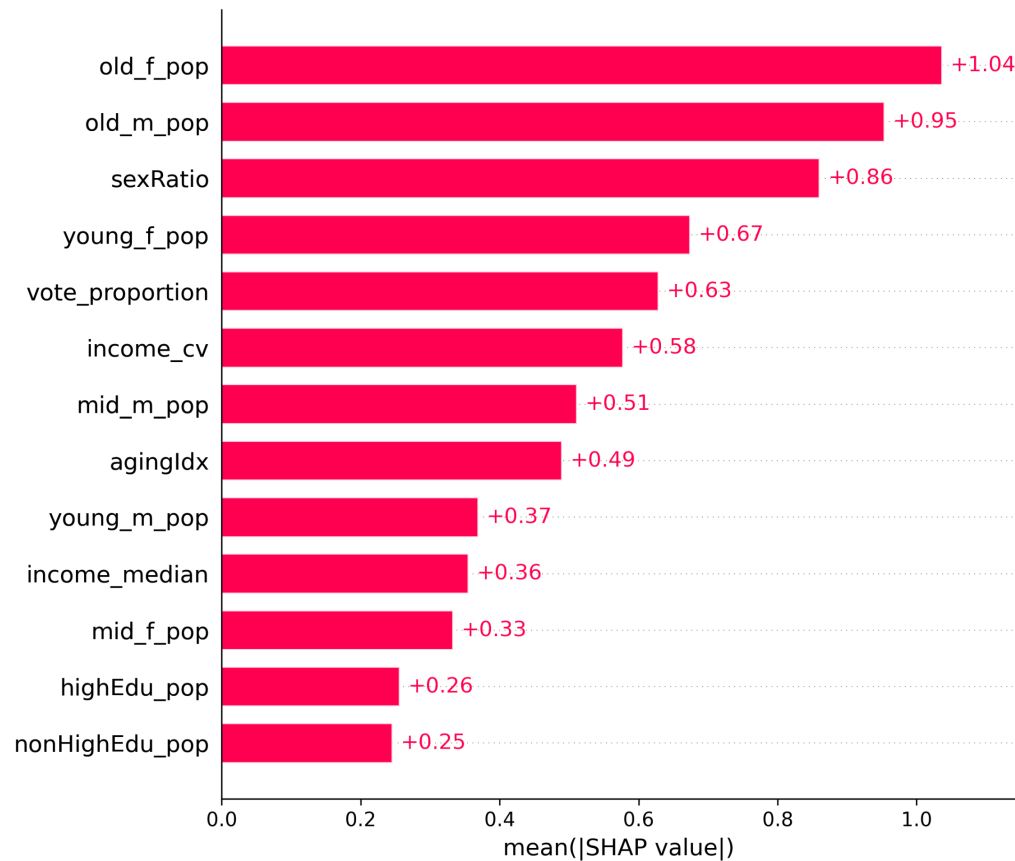
$$\hat{f}(\{A, B\}) = 0.45 - 0.30 = 0.15$$

Given A and B, the prediction increase 0.15 compared to baseline.

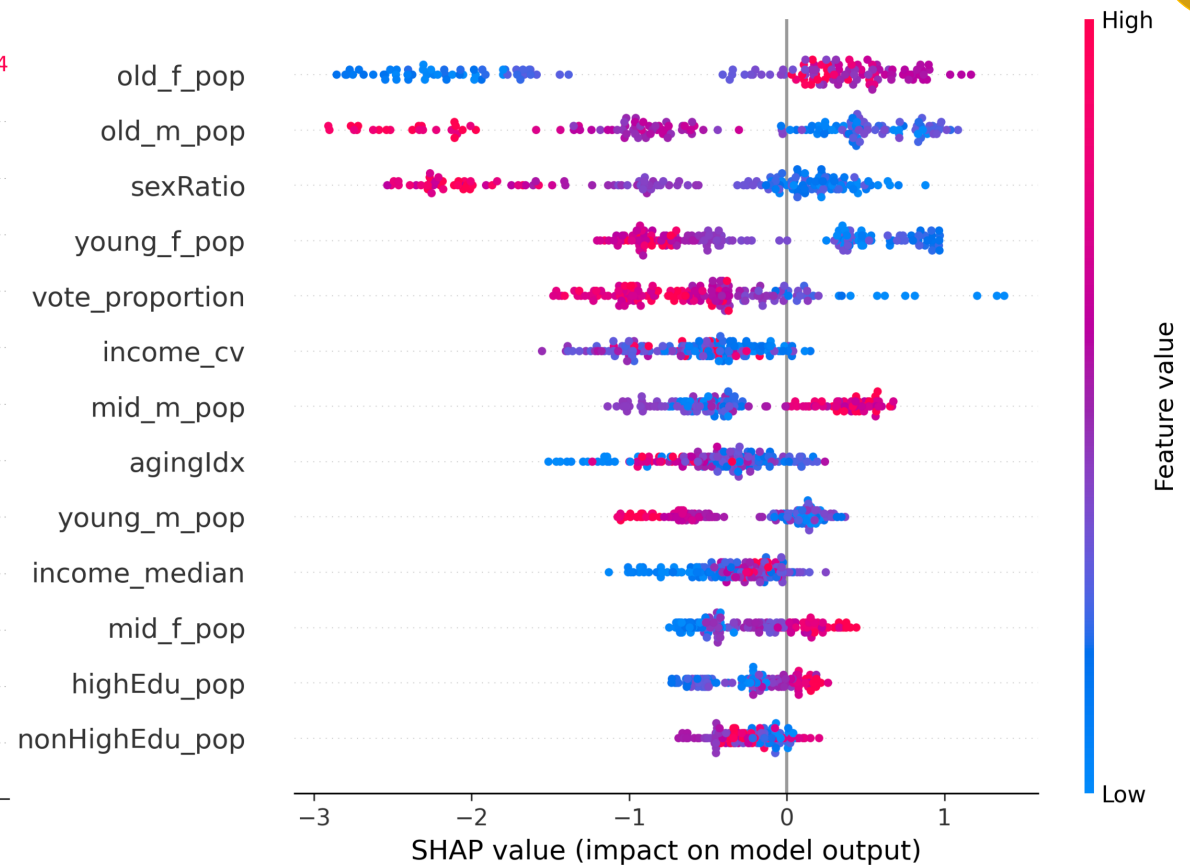
$$\hat{f}_x(S) = \text{prediction given features in } S - \text{baseline prediction}$$

SHapley Additive exPlanations

Bar Chart

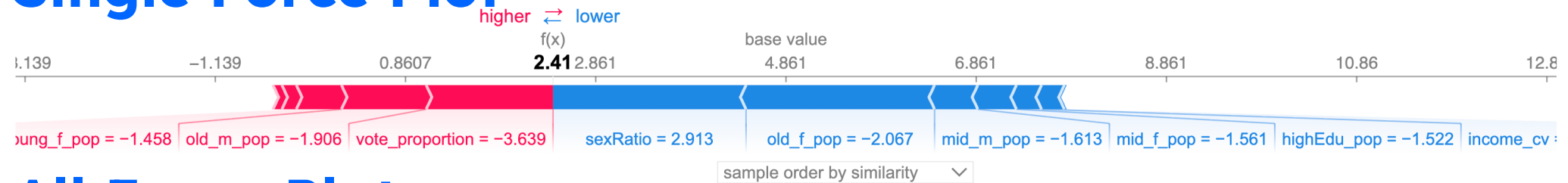


Beeswarm Plot

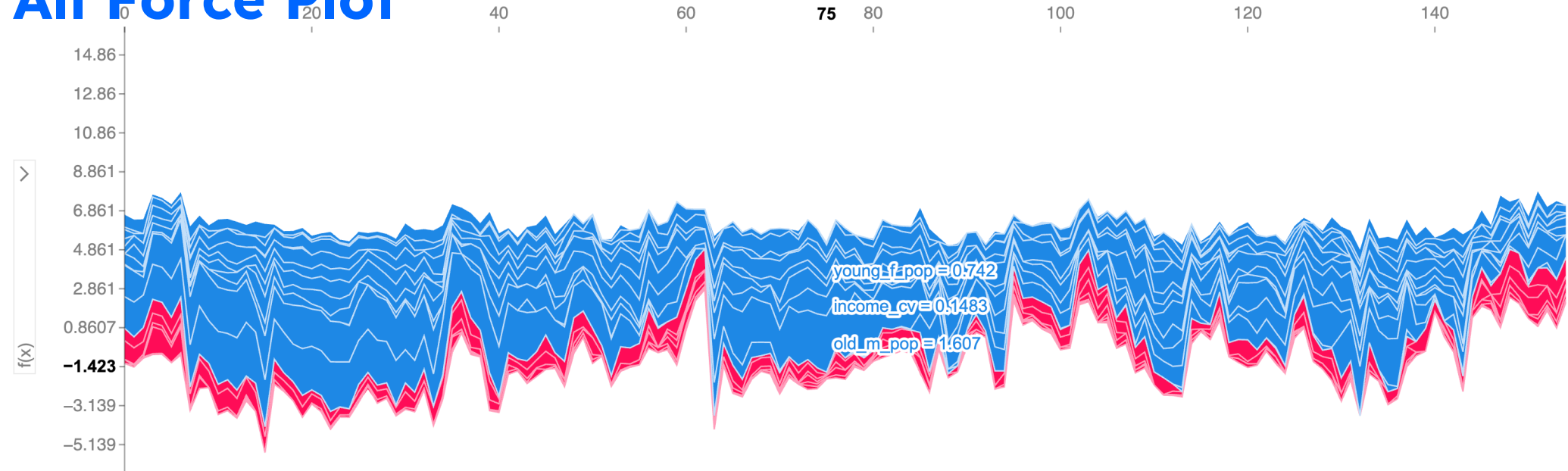


SHapley Additive exPlanations

Single Force Plot

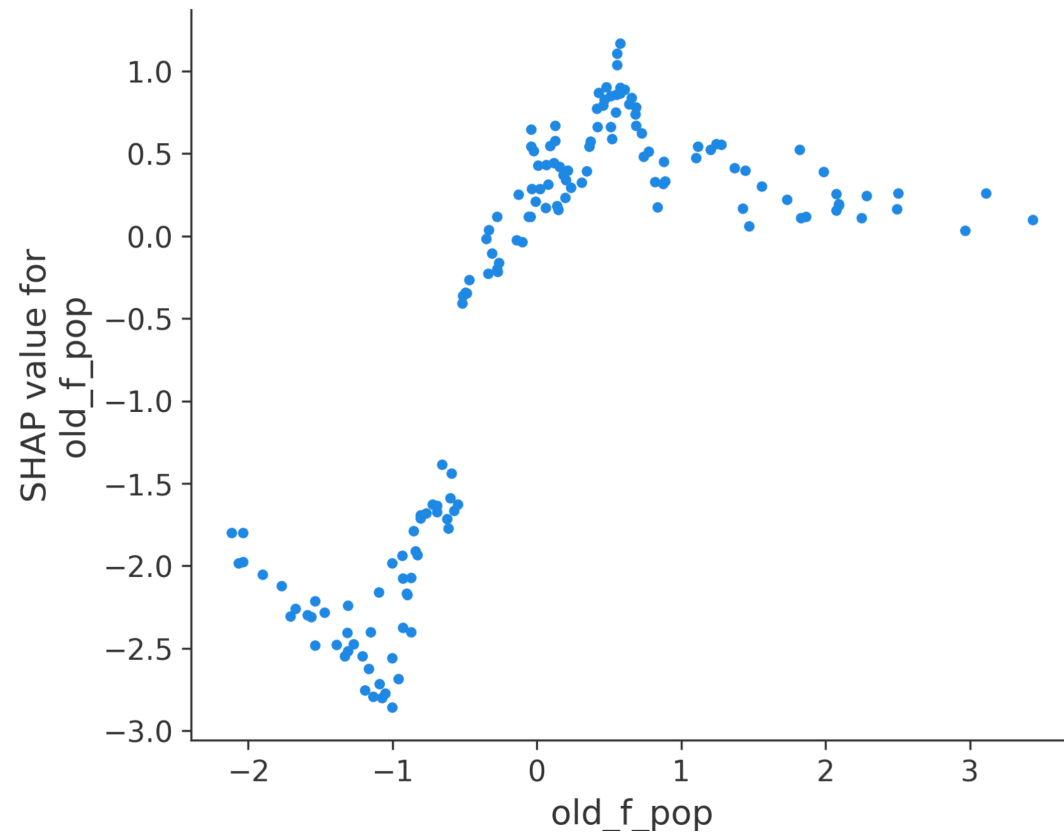


All Force Plot

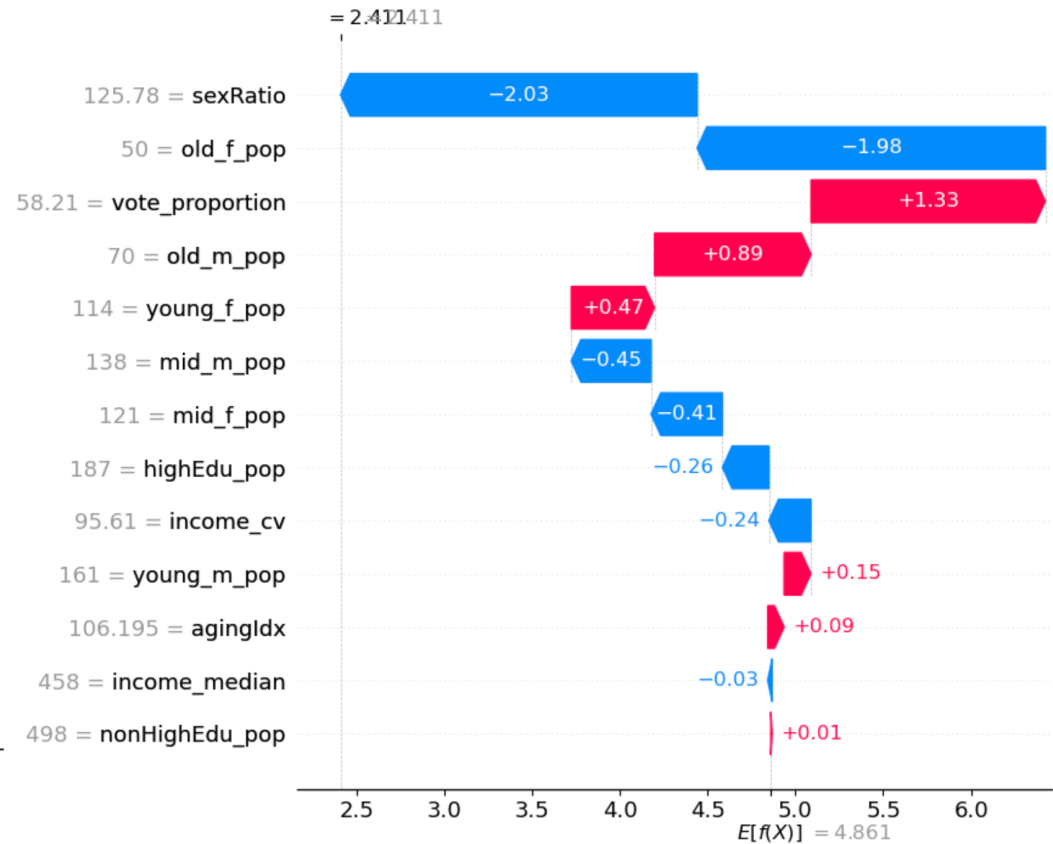


SHapley Additive exPlanations

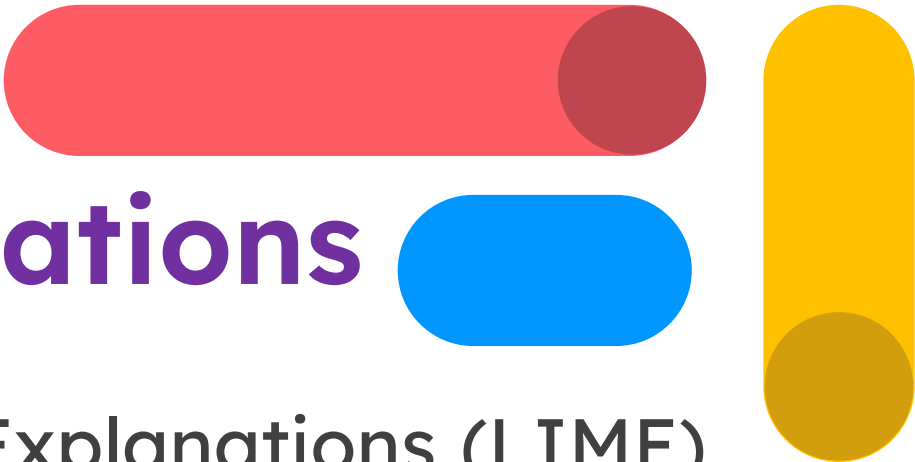
Dependence Plot



Waterfall



Local Interpretable Model-agnostic Explanations



Local Interpretable Model-agnostic Explanations (LIME) would like to know the reason why prediction probability is 0.82 for this patient, rather than asking the importance of a specific feature to the entire model.

Given a observation of $x_0 = (68, 31)$ for age and BMI, the model output for this model is

$$f(x_0) = 0.82$$

- 1) Generate perturbed samples around x_0
- 2) Give higher weights to nearby samples
- 3) Fit a simple interpretable model
- 4) Interpret the coefficients

Local Interpretable Model-agnostic Explanations

1) Generate perturbed samples around x_0

All generated perturbed samples are designed for testing the impact of each feature on the model output via the small changes of each feature.

Sample	Age	BMI	$f(z)$
z_1	67	31	0.80
z_2	69	32	0.85
z_3	63	29	0.68
z_4	55	25	0.40

Local Interpretable Model-agnostic Explanations

2) Give higher weights to nearby samples

To compute the proximity of each sample to the target sample, we conduct Euclidean distance.

$$d(z, z_0) = \sqrt{(z_{Age} - 68)^2 + (z_{BMI} - 31)^2}$$

Given $z_1 = (67, 31) \Rightarrow d1 = \sqrt{(67 - 68)^2 + (31 - 31)^2}$

You may calculate rest of them ...

Local Interpretable Model-agnostic Explanations

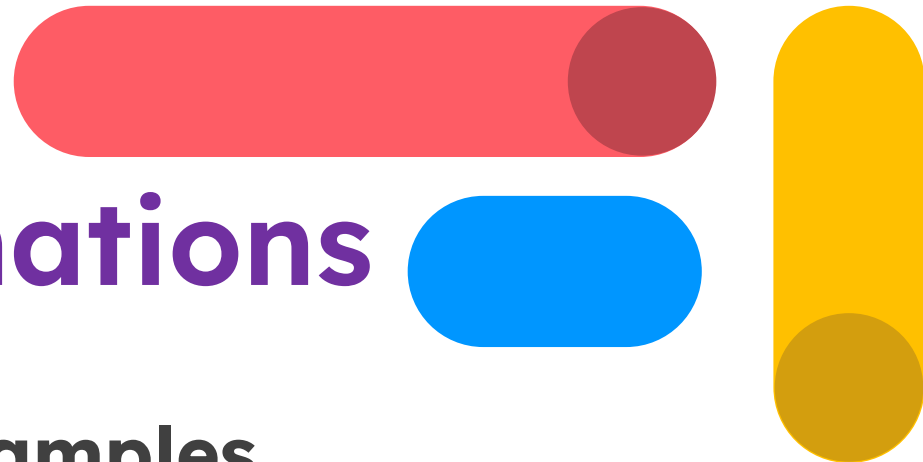
2) Give higher weights to nearby samples

To normalize the distance, here, we adopt kernel function to transform from original distance to weights. The nearer distance indicates the higher weight.

Due to the distance of z_1 , the penalty of incorrect prediction will seriously affect the model outcome.

Sample	Distance	Weight
z_1	1.0	0.90
z_2	1.4	0.82
z_3	5.4	0.25
z_4	14.3	0.01

Local Interpretable Model-agnostic Explanations



2) Give higher weights to nearby samples

The common approach to transform from distance to weight (importance).

$$weight = \pi_{x_0}(z) = \exp\left(-\frac{d(x_0, z)^2}{\sigma^2}\right)$$

Local Interpretable Model-agnostic Explanations

3) Fit a simple interpretable model

We have Age, BMI, model outcome ($f(z)$), weight of each sample (w_i), and now, LIME adopts a linear model.

$$g(z) = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{BMI}$$

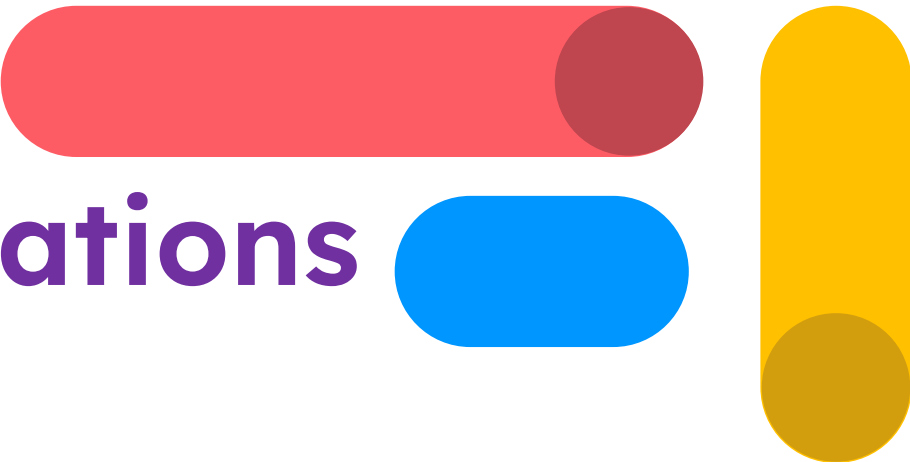
The target is $f(z)$.

The loss function of LIME is

$$\sum_i w_i [f(z_i) - g(z_i)]^2$$

Age	BMI	Target for g	Weight
67	31	0.80	0.90
69	32	0.85	0.82
63	29	0.68	0.25
55	25	0.40	0.01

Local Interpretable Model-agnostic Explanations

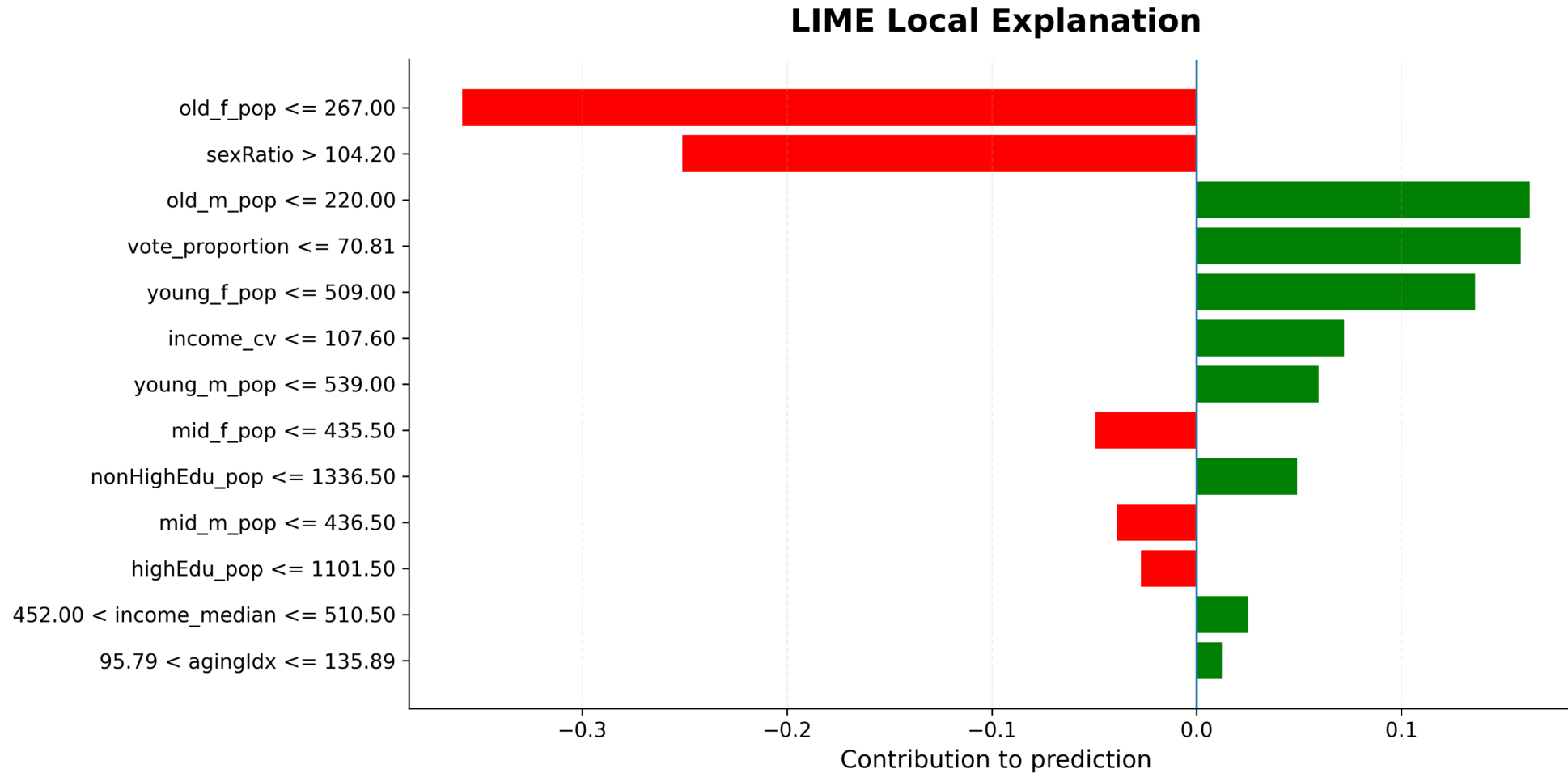


4) Interpret the coefficients

$$g(z) = \beta_0 + \beta_1 Age + \beta_2 BMI$$

The higher coefficients indicate the higher and positive impact of this feature on the model output.

Local Interpretable Model-agnostic Explanations

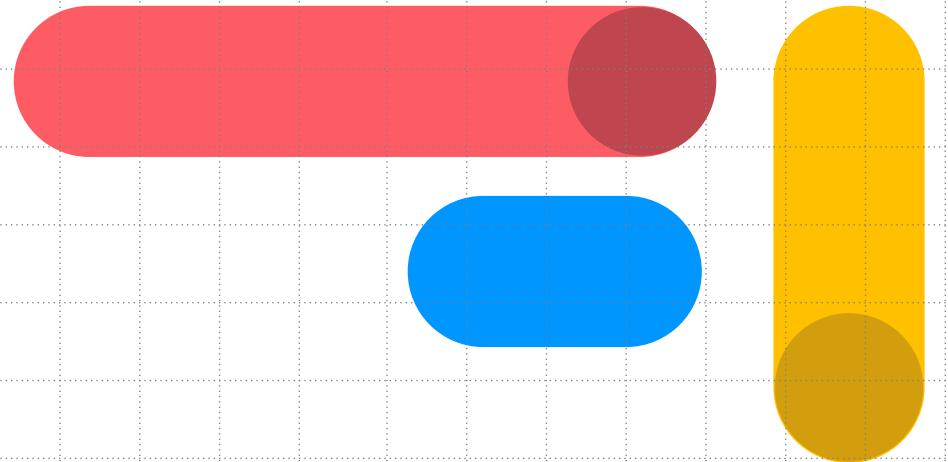


Summary

Method	Explanation Level	Explanation Type	Main Output	Direction/Effect	Model-Agnostic
Feature Importance	Global	Importance	Importance score for each feature	Usually No	Depends on the method
PDP	Global	Effect	Average prediction across feature values	Yes	Yes
ICE	Individual/ Local	Effect	One response curve per observation	Yes	Yes
SHAP	Local, aggregable to Global	Attribution	SHAP value for each feature	Yes	Yes, in principle
LIME	Local	Local Importance/ Surrogate Explanation	Local feature weights from a simple surrogate model	Yes, locally	Yes

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The End

Thank you for your attention!

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